

AR 501/ ME 452: Robot Kinematics, Dynamics, and Control

Course Code	AR501/ ME452
Course Name	Robot Kinematics, Dynamics, and Control
L-T-P-C	3-1-0-4
Prerequisites	Consent of faculty advisor
Intended for	UG, PG and Ph.D
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- **Introduction to Robotics:** Basic definitions, mechanism, degree of freedom, Rigid Body Motions - Fundamentals, Classification of robots, actuators, sensors, and control systems. (3 hours)
- **Kinematics:** Tracking Rigid Bodies (position and orientation), Coordinate transformation, Differential Kinematics, Kinematic Chains - Forward Kinematics (FK), Inverse Kinematics (IK), Differential Manipulator Kinematics. (11 hours)
- **Dynamics:** Rigid Body Dynamics - Dynamics of Constrained Particles, Dynamics of a Rigid Body; Manipulator Dynamics - Dynamics of Serial Manipulators, Manipulator Dynamics with Constraints. (11 hours)
- **Trajectory generation:** Determining the joint variables for desired trajectory. (5 hours)
- **Control:** Fundamentals of Control - Linear Time Invariant Systems with Single Input and Output, Feedback Control and Stability, PID Controller, State Estimation in Feedback Systems; Manipulator Control - Local vs Centralized Motion Control Strategies, Indirect vs Direct Force Control Strategies. (12 hours)

Laboratory Modules

Kinematics, Dynamics, Trajectory generation, Control

Text Books

1. Kevin M. Lynch and Frank C. Park, **Modern Robotics: Mechanics, Planning, and Control**, Cambridge University Press, 2017.
2. Craig John J., **Introduction to robotics: Mechanics & Control**, 3rd Edition, Pearson, 2008.
3. M. W. Spong and M. Vidyasagar, **Robot Dynamics and Control**, John Wiley, 1989.
4. Murray R., Li Z., and Sastry S., **A Mathematical Introduction to Robotic Manipulation**, CRC Press.
5. Siciliano B., Sciavicco L., Villani L. and Oriolo G., **Robotics: Modeling, Planning and Control**, Springer.

References

1. Ellis G., **Control System Design Guide**, Elsevier.
2. Jazar R. N., **Theory of Applied Robotics: Kinematics, Dynamics, and Control**, Springer.
3. Moon F., **Applied Dynamics**, Wiley-VCH.
4. Astrom K. and Murray R., **Feedback Systems: An Introduction for Scientists and Engineers**, Princeton.
5. Friedland B., **Control System Design: An Introduction to State-Space Methods**, Dover.